Ability concealment: scientific claims by category

An evidence-based assessment of expected demonstrations, literature-anticipated findings, and surprising model results.

29 September 2026 | E01-E05 evidence review

The claims concern the specified computational models and tested designs. The categories assess scientific interest; they do not rank certainty. "Post-hoc trivial" means understandable once the mechanism is considered, not an unplanned analysis.

1. Expected once the mechanism is considered

Claims	Demonstration and interpretation
C01-C02	**Weighting negative outcomes more strongly increases masking.** At fixed receptivity, masking rose by 40.3 percentage points. The direction and subjective-threshold shift follow from the reward transform.
C03	**Initial rejection reduces subsequent sampling.** The hot-stove mechanism lowers an action's learned value and restricts corrective experience [1].
C04-C05	**Corrective information reduces avoidance.** Forced exposure and counterfactual feedback reduced masking by 4.66 and 11.81 points. Their tested ordering is a useful treatment comparison.
C07-C08	**Linear payoffs retain endpoint optima; lower disclosure lowers exposure.** The exact reward and variance formulas explain these properties of the graded model.
C10-C11	**Recovery improves incentives; established acceptance favours authenticity.** Escape reduces future costs. In Accepted, identical transitions leave an exact authenticity advantage of a+c=.85.
C12-C13	**Dynamic rankings can differ from immediate rankings, and finite learning can miss the optimum.** Both are familiar consequences of future rewards and incomplete information.
C18	**Final success does not erase costs accumulated during search.** Endpoint classification and reward collected along the way measure different outcomes.
C22-C23	**Higher comparison cost can raise masking; similar behaviour can have different causes.** The contribution is the quantitative demonstration and explicit mechanism decomposition.

Role in the paper: establish the model's logic, quantify familiar mechanisms, and provide the controls needed to interpret the stronger results.

2. Expected from gifted-children research

The literature anticipates the qualitative direction. The model contributes explicit mechanisms, controlled comparisons, and numerical consequences.

C12 AND C23: SOCIAL INTERPRETATION

Concealment can serve a social purpose.

Information management and social coping already describe gifted students reducing visibility to manage acceptance and social consequences [2-4].

What the model adds. The model supplies an explicit distinction between dynamically adaptive concealment and masking that sacrifices available objective value.

C08; ENGAGEMENT COMPONENT OF C06

Visibility management can be selective or partial.

Hiding, conformity, helping, and selective presentation show that coping is richer than an all-or-nothing choice [2-4].

What the model adds. The graded model formalizes a repertoire of actions. The combination of higher engagement and lower endpoint use is the surprising extension.

C19; BROAD SOCIAL INTERPRETATION OF C11

Receptive peers and settings should facilitate expression.

Context-sensitive accounts of gifted social experience motivate this expectation [2,5].

What the model adds. Varying receptive-context abundance quantifies how the distribution of favourable opportunities changes learned behaviour.

C20: COMPARISON-COST INGREDIENT

High-ability settings can bring comparison costs alongside benefits.

The big-fish-little-pond literature reports reference-group effects on academic self-concept, anxiety, and performance [6].

What the model adds. The model represents comparison cost through a payoff parameter and computes its objective boundary. The learned-boundary displacement is a separate result.

C23: CONTEXTUAL INTERPRETATION

Visibility management varies across learners and settings.

The giftedness literature documents varied social experiences and coping responses [3-5].

What the model adds. The framework specifies distinct sources of variation and the observations needed to distinguish them.

Overlap is intentional: a familiar mathematical mechanism can also express a literature-anticipated social idea. A greater negative-outcome weight is a general learner parameter here, not an established characteristic of gifted children.

3. Surprising results

Strongest applied contrasts

A / C06

More gradual options preserve engagement while reducing fuller expression.

Independent graded actions increased post-rejection nonzero disclosure by **25.50 percentage points**, reduced full-authentic final-window selection by **32.87 points**, and increased normalized objective regret by **.02976**. The best achievable objective payoff was unchanged.

Why it is interesting. A plausible expectation is that intermediate steps both sustain participation and support progress toward fuller expression. The experiment separates those outcomes: engagement improves while intensity and endpoint use decline.

B / C15-C16

An unchanged opportunity becomes much less usable under some representations.

Expanding from 1 to 16 contexts with one receptive context increased masking from **26.36% to 84.19%** in the flat architecture and from **26.45% to 87.27%** in the hierarchical architecture. Under global-mask representation, masking changed from **27.23% to 25.77%**.

Why it is interesting. Some burden from added alternatives is expected. The strong difference between action representations is the more interesting result. The best full-information reward remains fixed throughout.

C / C17

Successful encounter and sustained use are sharply separated.

At 16 contexts, **99.72%** of flat learners and **99.64%** of hierarchical learners had experienced accepted authenticity in the receptive context. Final-window masking nevertheless averaged **84.19%** and **87.27%**, respectively.

Why it is interesting. Extensive later masking coexists with almost universal successful exposure. The result distinguishes encountering receptivity from learning its value and consistently acting on it.

Editorial priority

These three contrasts deserve prominent treatment: the engagement-endpoint trade-off, representation dependence, and the gap between successful encounter and sustained use.

3. Surprising results

Boundary displacement and persistent learning gaps

D / C21

The learned population boundary lies inside the objectively favourable region.

The flat K=4 good-authentic modal-classification curve crossed .5 at comparison cost **.312852** (95% bootstrap CI **.307836-.316918**), while objective indifference is **.470588**. Authenticity still held an expected-reward advantage of **.134076** at the learned crossing. The learner used **lambda=1**.

Why it is interesting. The resolved separation occurs without enhanced negative-outcome weighting. The crossing measures the fraction of learners whose most frequent individual action is good-context authenticity; it is distinct from a 50% total masking rate.

E / C09; C12 examples

Persistent social consequences substantially relocate the objective threshold.

The receptivity threshold for optimal authenticity shifts from the static value **.7167** to approximately **.384, .339, or .269** at escape rates 0, .2, or .8. At one verified setting, authenticity has immediate expected reward **-.20** versus **+.75** for masking, yet is dynamically optimal.

Why it is interesting. Dynamic reversal is a familiar principle. Its substantial extent and concrete social-state consequences are the useful surprise. These are deterministic objective benchmarks.

F / C14

Suboptimal masking survives substantial additional opportunity to learn.

At all five prespecified disagreement sentinels, the masking-optimality gap persisted in all five master replications through **5,000 steps**. It also survived the named population, window, weighting, and initialization changes. Mean Neutral-state masking at the extended-horizon sentinels remained about **90%-99.7%**.

Why it is interesting. Finite-learning error is expected. Its persistence through the executed sensitivity checks is more informative than demonstrating an initial oracle gap. The perturbations were tested separately.

These results are surprising model consequences, not a claim that their general principles are unknown in learning theory. Their scientific value lies in the controlled contrasts, resolved magnitudes, and retained effects.

Complete claim index

One primary assignment per claim; social interpretations can overlap categories.

ID	Category	Scientific claim
C01	1	Learner weighting produces large changes with the environment fixed.
C02	1	Weighting moves subjective indifference independently of the objective boundary.
C03	1	Initial rejection reduces later authentic sampling.
C04	1	Corrective information reduces masking and objective loss.
C05	1	The tested counterfactual treatment produces the larger reduction.
C06	3	Graded options increase engagement but reduce intensity and endpoint use.
C07	1	An affine graded payoff preserves the endpoint optimum.
C08	1	Partial disclosure lowers exposure and variance in favourable E02 cells.
C09	3	Persistent social state substantially relocates the objective threshold.
C10	1	Recoverability changes the objective incentive to disclose.
C11	1	Authenticity strictly dominates in the Accepted state.
C12	1	Future consequences can reverse either immediate payoff ranking.
C13	1	Finite learning can sustain objectively suboptimal action allocation.
C14	3	The masking-optimality gap survives extended-horizon sensitivity checks.
C15	3	Representation determines how usable an unchanged opportunity becomes.
C16	3	Context expansion sharply increases masking in the tested search architectures.
C17	3	Successful encounter does not guarantee sustained use.
C18	1	Search has payoff costs even when final modal selection is high.
C19	2	More widespread receptivity restores good-authentic modal selection.
C20	2	Comparison cost changes the objective appeal of a receptive setting.
C21	3	The learned modal boundary lies below objective indifference.
C22	1	Actual masking rises while authenticity retains objective advantage.
C23	1	Distinct mechanisms can generate similar observed concealment.

Primary counts: 14 expected demonstrations; 2 primarily literature-anticipated claims; 7 surprising claims, forming six themes. These are editorial assignments, not independent-discovery counts.

Evidence and references

Evidence scope

E01, E02, E04, and E05 standard cells used 500 learners, 2,000 steps, a 400-step final window, and five independent master seeds. E03 baseline cells used 160 learners, 1,000 steps, a 200-step window, and five masters. E03 robustness statements refer to the named prespecified sentinel checks.

Scalar confidence intervals use master-seed estimates. E05 crossing uncertainty uses a bootstrap of complete master curves. Additional rates reported here are archived secondary outcomes or descriptive means, as documented in the source review. The E01-E03 historical generator-source limitation remains recorded in the evidence documentation.

Research references

[1] Denrell, J., & March, J. G. (2001). Adaptation as information restriction: The hot stove effect. *Organization Science, 12*(5), 523-538.
https://doi.org/10.1287/orsc.12.5.523.10092

[2] Coleman, L. J., & Cross, T. L. (1988). Is being gifted a social handicap? *Journal for the Education of the Gifted, 11*(4), 41-56.
https://doi.org/10.1177/016235328801100406

[3] Swiatek, M. A. (1995). An empirical investigation of the social coping strategies used by gifted adolescents. *Gifted Child Quarterly, 39*(3), 154-160.
https://doi.org/10.1177/001698629503900305

[4] Klimecká, E. (2023). Advantages and disadvantages of being "gifted": Perceptions of the label by gifted pupils. *Research Papers in Education, 38*(6), 902-923.
https://doi.org/10.1080/02671522.2022.2065523

[5] Dublin City University, Centre for Talented Youth Ireland. *Irish Gifted Students: Self, Social, and Academic Explorations.* Research report and findings.
Research report - DCU / CTYI

[6] Zeidner, M., & Schleyer, E. J. (1999). The big-fish-little-pond effect for academic self-concept, test anxiety, and school grades in gifted children. *Contemporary Educational Psychology, 24*(4), 305-329.
https://doi.org/10.1006/ceps.1998.0985

Document sources and wider model ladder

Prepared from the 29 September 2026 Supported Scientific Claims review (C01-C23), Scientific Contribution Categories, the E01-E05 evidence matrix, archived result tables, and analytic-results authority. Claim IDs preserve the connection to those records. The targeted literature check establishes context; it is not an exhaustive novelty search.

The wider M4 model's geometric recurrence, conditional drift, and mean-drift roots are useful exact results in category 1. They remain outside this document's E01-E05 empirical claim set.